VF Asia Footwear Laboratory Testing Manual

Test Method for Water Vapor Permeability and Absorption	**Test method # VF- E25** **REF # : TM-R-053** **Page 1 of 3**
Date Issued : September 12, 2014 **Revision # :**	**Created by : Nila Babia Erazo/ Chris Adam**

PURPOSE AND SCOPE: To determine the amount of water vapor an assembly or single material will absorb and transmit through its structure in a specified time.

REFERENCE: SATRA Test Method TM 47, "Water Vapor Permeability and Absorption," April 1997.

SATRA Test Method TM 172, "Water Vapor Permeability."

SATRA Test Method TM 178, "Water Vapor Absorption."

EQUIPMENT AND MATERIAL:

1. Moisture Vapor Transfer Rate (MVTR) test apparatus in compliance with SATRA 147.
2. Analytical balance capable of measuring mass to the nearest 1mg.
3. A thermometer capable of measuring temperature to the nearest 1°C.
4. Distilled or deionized water.
5. 54mm (2.125 in) diameter clicker die.
6. Clicker press.
7. Hot melt glue gun.
8. Footwear component

SAMPLE PREPARATION:

1. Condition the sample for at least 48 hours at standard laboratory environment at 23.0±2.0 ˚ C temperature and 55 ±10 % Relative Humidity.
2. Die or hand cut a test sample of 54mm diameter test specimens.
3. Uniquely label each test specimen.

REPETITIONS: Test two (2) samples per material**.**

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0 ˚C temperature and 55 ±10 % Relative Humidity.

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1. Weigh each test specimen to the nearest mg. Record these values as [Mo]
2. Seal the edges of each test specimen with hot melt glue.
3. Weigh each test specimen to the nearest mg. Record these values as [Ms]
4. Turn on the test apparatus described in section 2.1 and allow it to reach a temperature of 32 ± 2°C.
5. Pour warm distilled water, temperature approximately 35°C, into each of the pots to a depth of approximately 5mm.
6. Place a test specimen centrally over the open end of each pot such that the surface that would be closer to the foot in the final product is lowermost. For example the face of an upper material should be uppermost, whereas the face of a lining material should be lowermost.
7. For each pot place a clamping ring centrally over the top of the specimen assembly and tighten the ring so that the assembly is securely held around its edge and the pot is sealed.
8. Measure the weight of the combined pot and specimen assembly to the nearest mg and record this weight as [Mp0]. Take care to keep the pots vertical and avoid rapid or jerky movement which might splash distilled water on the test specimen assembly.
9. Place each of the test pots into the test apparatus.
10. After 60 ± 5 minutes carefully remove the test pot assemblies from the test apparatus. Again, take care to keep the pots vertical and avoid rapid or jerky movement which might splash distilled water on the test specimen assembly.
11. Dry the base of the pot assemblies, measure the weight and record this weight as [Mp1] to the nearest mg.
12. Repeat steps 10 and 11 after elapsed times of 2, 3, and 4 hours. Record the weights as [Mp2], [Mp3], and [Mp4] respectively.
13. After a total elapsed time of 265 ± 5 minutes repeat steps 10 and 11 Record this weight as [Mp265]
14. Immediately dismantle the pot and specimen assembly taking care not to splash any of the water onto the test specimen assembly.
15. Weigh each test specimen to the nearest mg. Record these values as [M265]. Carefully re-assemble the test specimen assemblies on the pots and return them to the test apparatus.
16. After elapsed times of approximately 5 and 6 hours repeat steps 10 and 11. Record these weights as [Mp5] and [Mp6] respectively.

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17. After an elapsed time of 400 ± 5 minutes, repeat the procedure in steps 10 and 11 Record the weight as [Mp400]. Then repeat the procedure in steps 14 and 15. Record the weight as [M400], but do not re-assemble the test specimen assemblies.

CALCULATION:

Calculate the water vapor permeability (WVP) and water vapor absorption (WVA), after 265 minutes and 400 minutes, for each test specimen according to the procedure outlined in Ref. 1.1, and using the Timberland MVTR spreadsheet.

As an alternative, SATRA Test Methods TM 172, “Water Vapor Permeability” (Ref. 1.2) and TM 178, “Water Vapor Absorption” (Ref. 1.3) may be used to determine water vapor permeability and absorption.

REPORT:

1. WVP,
2. WVA 265,
3. WVA 400,

NOTE: Also report material description, any deviations from this procedure, and SATRA test method used for the determination of these properties